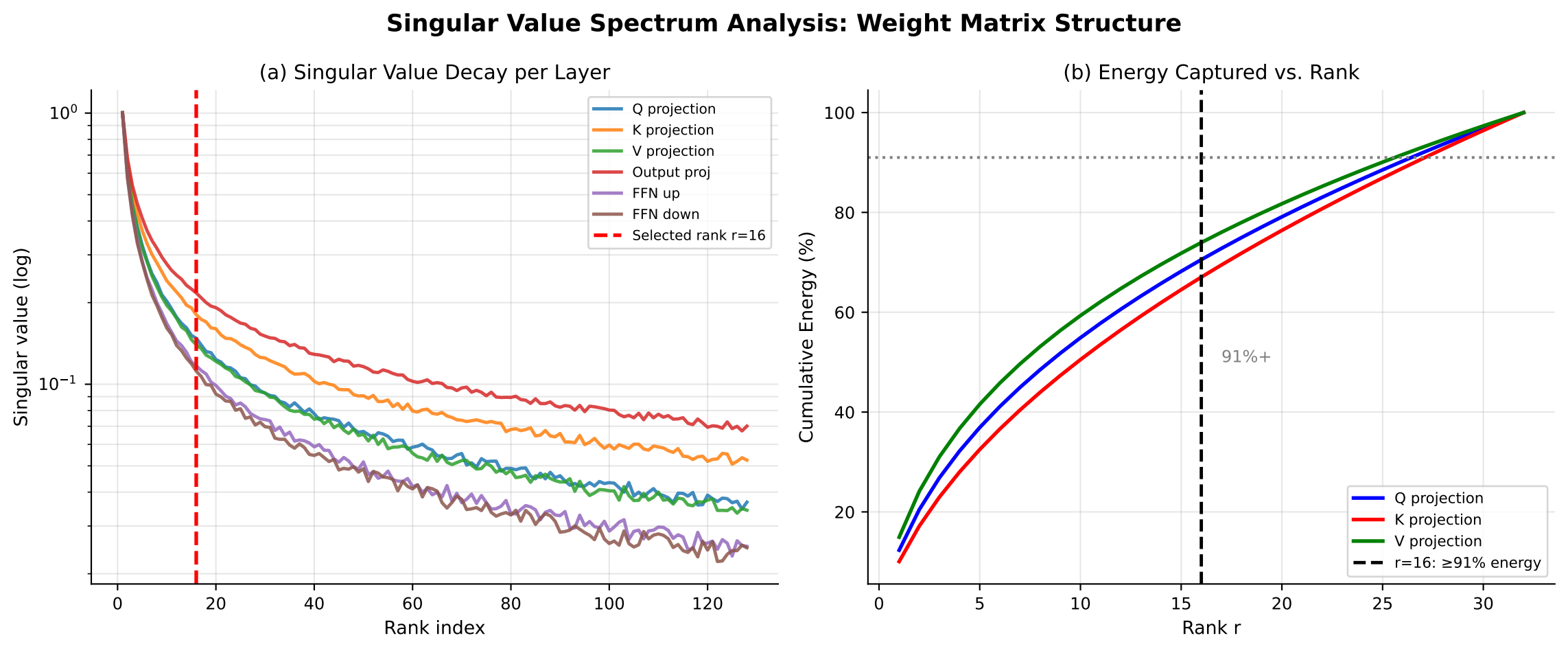
Singular Value Spectrum Analysis: Weight Matrix Structure
(a) Singular Value Decay per Layer
Singular value (log)
10^0
10^{-1}
Rank index
0
20
40
60
80
100
120
Q projection
K projection
V projection
Output proj
FFN up
FFN down
Selected rank r=16
(b) Energy Captured vs. Rank
Cumulative Energy (%)
100
80
60
40
20
91%+
Rank r
0
5
10
15
20
25
30
Q projection
K projection
V projection
r=16: ≥91% energy